

Integrating Post-Quantum Cryptography: A New Frontier in Cyber Defense and Threats

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Abstract -

Quantum computing is progressing at an alarming rate, threatening traditional cryptography; for example, widely used encryption methods like RSA and ECC may soon be rendered ineffective. Leveraging the high degrees of parallelism offered by qubits and algorithms such as Shor's, quantum computers have the potential to break encrypted information that is presently secure, raising a serious threat to data confidentiality and national security. Therefore, post-quantum cryptography (PQC) has emerged as a proactive defense strategy. However, PQC itself is a double-edged sword. It is intended to protect digital communications from quantum-powered attacks, yet adversaries could also utilize harmful objectives for its deployment and functionalities. This paper presents evidence on the real-world implementation of PQC through a succession of case studies across such key sectors as banking, health care, and education. We examine each of these sectors for their unique vulnerabilities, their implementation challenges, and the extent to which PQC solutions can effectively address them. In addition, the paper discusses the attacks against the post-quantum algorithms, both on theoretical and practical evidence of attacks. It will describe how the adversary utilizes transitional security flaws to gain an advantage in the post-quantum world. It further covers the tools, protocols, and frameworks created for the deployment of

PQC, analyzing their performance and security assurances. This study aims to assess where PQC stands and where it is heading. It also aims to guide the development of cybersecurity measures and informed policies around PQC adoption by evaluating the pros and cons of PQC implementation and its wider impact on the cybersecurity landscape and the burgeoning quantum-enhanced society.

Key terms: Post-Quantum Cryptography, Quantum Computing, Cybersecurity, Cryptographic Protocols, Banking, Healthcare, Education, Cyber Attacks

I. Introduction

Cryptography is the science of securing communication and has been around for thousands of years, dating back to methods such as the one used by the Romans called the Caesar cipher. Over the 20th century, modern cryptography started evolving, particularly with the arrival of computers. The code-breaking of the German Enigma cipher during World War II, particularly by Alan Turing, was a critical turning point in the war for the Allies. The advent of public-key cryptography in the 1970s, marked by our discovery of the RSA algorithm, laid the foundation for secure key exchange over insecure channels without revealing shared secrets. Since then, cryptographic protocols have become ubiquitous. They can be used to secure Internet communications, facilitate online

transactions, and verify digital identity (Diffie & Hellman, 1976).

Quantum computers are a new paradigm for computation, using properties like superposition and entanglement to perform certain classes of computations vastly more efficiently than classical computers. However, in 1994, Shor proposed a quantum algorithm, AKA Shor's algorithm, which can factor large integers and compute discrete logarithms efficiently in polynomial time (Shor (1994). Though still in the experimental stage, sufficiently powerful quantum computers would theoretically be capable of executing algorithms like Shor's, capable of cracking classical encryption schemes, most famously RSA and ECC, that underpin everything from financial systems to national defense networks. This ability poses a severe cybersecurity risk for quantum computers. National Academies of Sciences, Engineering, and Medicine (2019) states that all current public-key cryptography will be endangered after the construction of large-scale quantum computers since it will undermine the confidentiality, integrity, and authenticity of digital information.

Quantum computers threaten classical cryptography because they would break the mathematical foundations on which the existing systems are built. Encryption algorithms like RSA and ECC are still secure; however, their security originates from factorization and solving discrete logarithms, which a classical machine could never solve practically. On the other hand, a Quantum Machine can solve it in a polynomial amount of time by applying quantum Algorithms. This was a major vulnerability as quantum attackers might break digital signatures, decrypt past communications, or forge certificates. This has led to urgent efforts towards quantum-safe systems (Mosca, 2018).

Post-quantum cryptography (PQC) was born from the awareness that standard cryptographic algorithms could be cracked as quantum computer power increases.

Interest in PQC was piqued by the discovery of Shor's algorithm in 1994. In 2016, NIST began a public competition to standardize quantum-resistant algorithms. That project led to the selection of several finalists in 2022, including Crystals Kyber for key exchange, and Crystals Dilithium and Falcon for digital signatures. These employ challenging problems based on lattice structures and hash functions and are in the process of being finalized for standardization (Chen et al., 2016).

Post-quantum cryptography (PQC) refers to cryptographic algorithms considered to be secure against quantum attacks. Unlike classical systems that depend on number theoretic problems, PQC methods employ the hardness of mathematical problems based on lattices, code decoding, or multivariate polynomials. Long-term data confidentiality and trust in digital systems depend on post-quantum cryptography (PQC). To prepare for the quantum era, organizations and governments around the world are starting to incorporate PQC into their safety infrastructure (Campagna et al., 2023).

This paper describes both parts of Post-Quantum Cryptography (PQC), a technology that can strengthen the defense against quantum computing attacks and introduce new attack vectors. PQC seeks to fortify sensitive systems against quantum attacks that threaten to disrupt and compromise secure communications, transactions, and data storage. In contrast, deploying immature or poorly implemented PQC can introduce additional vulnerabilities, affect performance, and increase the attack surface. This paper will explore these tradeoffs and consider how organizations can position themselves for a successful and safe transition.

II. Understanding Post-Quantum Cryptography (PQC)

Post-quantum cryptography (PQC) is a cryptography that helps to develop algorithms that can be safe in front of

powerful quantum computers. Although the present cryptographic standards, including RSA (Rivest–Shamir–Adleman) and ECC (Elliptic Curve Cryptography), are secure against classical computing-based attacks, they are primarily vulnerable to quantum algorithms, such as Shor’s algorithm, which can practically factor large integers and can compute discrete logarithms efficiently. If quantum computers can reach the necessary scale and stability, they could break much of the public-key infrastructure on which all manner of things, from online banking to encrypted emails, have come to rely. PQC research has targeted complex mathematical challenges that do not yield readily to quantum approaches to meet this impending threat. These range from so-called lattice-based problems, code-based problems, hash-based schemes, multivariate polynomial equations, and isogeny-based constructions.

In contrast to quantum cryptography, which utilizes the laws of quantum physics for key distribution, PQC is purely classical. This means that deploying it onto existing hardware and software infrastructure is easier, making PQC a much more practical and scalable solution for organizations looking ahead to the quantum age in the shorter to medium term. For the development of post-quantum cryptographic standards, the National Institute of Standards and Technology (NIST) has been at the forefront, organizing a multi-round global competition to evaluate, test, and standardize quantum-resistant public key algorithms, which was initiated in 2016. With strong candidates like Kyber and Dilithium cutting, the world is preparing for post-quantum application security. These algorithms are the foundation of next-generation cryptographic standards, replacing weak algorithms before quantum computers become a practical threat. Moving to PQC is not just another tech upgrade train but a journey to prepare for a post-quantum world, where confidentiality, integrity, and authenticity in digital space will be a life as usual (Chen et al., 2016).

PQC algorithms rely on mathematically complex problems that are conjectured to be impractical for classical as well as quantum computers to solve reasonably, even with future advances in quantum computing technology. These issues are central to the cryptographic schemes intended to resist the power of quantum algorithms like Shor’s and Grover’s. One of the most promising families is lattice-based cryptography, which forms the basis of schemes like Crystals-Kyber (for key encapsulation) and Crystals-Dilithium (for digital signatures). Not only are lattice-based algorithms now widely believed to be highly resistant to quantum attacks, but they are also computationally efficient, showing potential for integration in existing hardware and network protocols. They provide favorable performance characteristics, such as fast key generation, encryption, decryption, and small key sizes, in comparison with other post-quantum alternatives.

A more mature category is hash-based cryptography, represented by the Sphincs+ digital signature scheme. These algorithms depend only on cryptographic hash functions, which are easy to understand, well-studied, and exceedingly difficult to reverse-engineer, even with quantum. Hash-based signatures generally may have larger signatures and slower signing performance, but their conceptually strong security under weaker assumptions makes them attractive for high-assurance applications, such as firmware integrity and secure boot.

Apart from such lattice problems, code-based cryptography relies on the hardness of decoding general linear error-correcting codes. For example, the Classic McEliece algorithm is a problem that has survived decades of cryptanalysis. Both code-based schemes generally have huge public keys but are considered quite robust and resistant to quantum attacks; therefore, they are well-positioned for use now as a means of long-term encryption. All these PQC classes have their tradeoffs regarding key/signature sizes, computational cost, and

implementation complexity, so their appropriateness usually depends on the application context. New design paradigms minimize classical computing without compromising the key functionality raised (Bindel et al., 2022), and as the field matures, researchers and standards bodies strive to strike these tradeoffs to arrive at appropriate post-quantum solutions that are secure, efficient, and scalable.

In 2016, the National Institute of Standards and Technology (NIST) initiated an extensive, multi-phase competition to discover and standardize quantum-resilient cryptographic algorithms. This effort was driven by the emergent threat of quantum computers, capable of attacking classical systems implementing RSA and ECC, which are susceptible to Shor's algorithm. NIST is trying to find algorithms that are secure against all forms of attack, both classical and quantum, but also efficient, practical, and interoperable so they can be deployed widely and across many different applications. The standards created by this competition will be helpful to and utilized by organizations worldwide, enabling a smooth transition to post-quantum security in the coming years. Different criteria will be considered to evaluate algorithms, including key size, efficiency, and resistance against classical and quantum attacks.

In July 2022, NIST revealed four finalist candidates for post-quantum cryptographic standardization for public-key encryption and digital signatures. The Crystals Kyber, a lattice-based cryptographic algorithm approved for key encapsulation mechanism (KEMs), was chosen for its high degree of security with lattice-based cryptography and for achieving excellent performance security tradeoffs. In addition, Crystals-Dilithium, which is also lattice-based, was selected for digital signatures due to its efficiency and strong security assurances. Falcon is a signatures-only scheme based on short integer solutions to polynomial equations. It was selected for its provable security against quantum attacks because it relies only on the algebraic structure of these

problems and Sphincs+, a hash-based signature scheme. Thus, while Falcon is known for its minor key and signature sizes, Sphincs+ is recognized for its security based on cryptographic hash functions. These Euclidean selections are important steps toward ensuring that quantum-resistant systems will be practical and secure for use in the real world. Final approval of these algorithms and formal standardization is projected to be from 2024 to 2025, culminating in more than a decade of research, development, and testing. As the public and private sectors prepare for a future where quantum computing could significantly disrupt data security (Moody et al., 2022), this global initiative is important to ensure they are ready.

PQC algorithms are expected to provide stronger security against quantum computing attacks, but deploying these algorithms introduces different vulnerabilities and difficulties. Among the most notable tradeoffs are the enlarged key sizes and signature lengths required for many PQC algorithms. Lattice-based schemes such as Crystals Kyber and Crystals Dilithium require keys that tend to be much larger than those used in classical algorithms such as RSA or ECC. Key exchanges and signatures are larger, and this additional size can significantly burden bandwidth. Moreover, the computational cost associated with using larger keys and signatures can result in increased latency for systems that demand real-time or high throughput transactions, such as those used for secure online communication or digital money transfers.

Furthermore, the use of PQC algorithms opens up a fresh attack surface for wannabe hackers to exploit. Switching to new cryptographic systems is risky because organizations must modify their infrastructure and software to accommodate these algorithms. Flaws in implementing new cryptographic libraries and side-channel vulnerabilities are misconfigurations that could unintentionally leak sensitive information or introduce a backdoor. For instance, a side-channel

attack might take advantage of tiny physical differences in the execution of a cryptographic operation, such as differences in timing or variations in power consumption, that could allow the extraction of private keys or other sensitive information. In fact, during the transition period, legacy and quantum-resistant systems may exist simultaneously, creating possible hybrid vulnerabilities. Also, the absence of large-scale real-world testing on various PQC algorithms must be weighed in cold scenarios, which will cause many unforeseen problems to surface at scale in real-world production setups. New challenges focus on the paradox in PQC, arguably as these algorithms give strong safety over quantum threats. However, their specification may expose novel vulnerabilities that require to be paid proper attention. This transition is complex, so extensive testing, continual security reviews, and full deployment instructions must be used to ensure post-quantum mechanisms do not risk security by accident (Alkim et al., 2021).

III. Case Study 1: Banking Sector

A. Organization: JPMorgan Chase

JPMorgan Chase is one of the biggest financial firms in the world, which means it works with massive amounts of sensitive financial data and communication. Because it is a critical infrastructure organization, it is still a high-value target for cybercriminals.

B. Type of Attack: A large-scale data breach due to phishing that resulted in credential theft and unauthorized access

One of the most well-documented attacks was a data breach at JPMorgan Chase in 2014, marking one of the most significant security incidents in U.S. banking history. In this case, cybercriminals obtained their hands on the PII of 83 million customers, comprising 76 million households and another 7 million small businesses. The bank said the stolen data

included names, addresses, phone numbers, and email addresses but did not include financial account numbers, passwords, or Social Security numbers. Following phishing campaigns to collect employee credentials, the attackers executed a multi-layer attack that exploited misconfigured servers and a server that did not implement two-factor authentication.

This permitted the attackers to move laterally in JPMorgan's network, eventually compromising over 90 servers. The breach remained undiscovered for months and was only detected after outside cybersecurity companies noticed unusual activity. The scale and sophistication of the operation set off speculation that it was the work of a nation-state. Nevertheless, it was subsequently revealed that the attackers were members of a criminal syndicate who sought to turn a profit. The breach highlighted deep flaws in enterprise-level security. It triggered JPMorgan and other financial institutions' massive rewiring of cybersecurity strategies, from network segmentation to enhanced monitoring and encryption technologies. (Harmon, 2016, p. 32).

C. Attack Tools Used: Phishing, server misconfiguration, and credential theft

The platform launched with a phishing campaign targeting individuals within the company by masquerading as a trusted third party, whether internal or external to the company. Unwitting employees were baited into handing over valid login credentials, providing attackers with a foothold in the bank's internal network. That early success paved the way for a more serious compromise.

Despite making it into the system, the attackers deployed their full arsenal against a critical misconfiguration of a server; one critical server was not set up for two-factor authentication, which, in turn, a widely adopted security control was designed to prevent unauthorized access even when credentials have been stolen. Had this second layer of defense, the

attackers would have found it much easier to escalate their privileges, and lateral movement within the network would not have proceeded as smoothly. Over the past few months, they methodically hacked over 90 internal servers, stealthily avoiding detection by masquerading as only legitimate traffic on the network and playing low on their operational footprint. (Kshetri, 2016, p. 93)

The attackers eventually exfiltrated the personally identifiable information of more than 83 million individual and small business customers, including names, email addresses, and phone numbers. While no financial data or passwords were said to be compromised, the size and stealth of the attack exposed disturbing weaknesses in enterprise cyber security architecture. This breach highlighted how a single misconfigured asset and successful phishing and credential theft can bring down even the most advanced networks. It also emphasized the value of ongoing network monitoring, access restrictions, and employee cybersecurity education as critical lines of defense.

D. Security Tools Used: Encryption protocols, newer TLS versions, firewalls, and threat detection systems

Furthermore, in light of the 2014 data breach, JPMorgan Chase launched an extensive overhaul of its cyber security infrastructure to address future risks and restore customer confidence. The most important upgrade was the implementation of end-to-end encryption for data in transit between internal and external communication streams. In addition, the bank enhanced its Transport Layer Security (TLS) configurations, turning off older versions and weak cipher suites to mitigate the risk of protocol downgrade attacks and ensuring strong encryption for web-based services.

In addition, JPMorgan invested drastically in enterprise-grade firewalls, intrusion detection systems (IDS), and security information and event management

(SIEM) tools, which allow them to monitor, log, and respond to potential threats in real time. Also, beyond these technical tools, the company adopted multi-factor authentication (MFA) on internal systems and took new measures around employee security awareness training to reduce susceptibility to phishing. In line with best practices, JPMorgan also began adopting stricter compliance with NIST-recommended cryptographic standards to future-proof the systems against classical and quantum-based threats. The multi-layered strategy mirrors wider activity in the financial services sector, adopting a defense-in-depth strategy in response to high-profile breaches.

E. PQC Implications Harm and Help

The cryptographic roots of the secure internet communications we enjoy today, such as RSA and Elliptic Curve Cryptography (ECC), which form the basis of most TLS protocols, are incredibly vulnerable to attacks by sufficiently powerful quantum computers. If such machines are ever built, they could run Shor's algorithm to break these schemes efficiently, thus enabling the decryption of previously recorded sessions of TLS-encrypted traffic. This retroactive threat pressurizes data intercepted and stored today to become compromised in the long run, sparing sensitive financial information and private conversation. For high-value targets such as JPMorgan Chase, this risk highlights the urgency of quantum-resistant cryptographic upgrades to maintain confidentiality over time (Mosca, 2018).

Researchers and institutions are innovating Post-Quantum Cryptography (PQC) to counter these threats and safeguard communications from quantum attackers. One leading example is Crystals Kyber, a lattice-based key encapsulation mechanism that is one of the variants NIST selected as a standard candidate to replace vulnerable public key cryptosystems within protocols such as TLS. JPMorgan Chase is already testing quantum-safe

communication protocols and has a head start on protecting its digital infrastructure due to the nature of quantum computing. These initiatives indicate a crucial transition in the financial sector towards quantum-proof cryptographic safeguards that can preserve security even in a post-quantum world (Campagna et al., 2023).

IV. Case Study 2: Healthcare Sector

2021 Universal Health Services Ransomware Attack

Universal Health Services (UHS) is one of the largest healthcare providers in the United States., with facilities across the United States and the United Kingdom. In 2021, the UNS was attacked by a ransomware attack. This incident had a devastating impact on the company and its revenue. This attack was one of the most significant ransomware attacks. Not only was their digital infrastructure exposed, but the hospitals could not protect sensitive data. It was estimated that UnitedHealth Services lost \$ 67 million in revenue during the three weeks it took to recover.

Universal health services are responsible for over 500 hospitals and health care facilities across multiple counties. These hospitals hold data and records for millions of patients. In the fall of 2020, multiple healthcare facilities reported that computers were shutting down and displaying ransom messages on them. Different systems all over the hospitals were unable to be used because of the malware used in this attack. It's believed that the malware used for this attack was Ryuk ransomware. Ryuk is a disruptive malware infamous for being used against large organizations. Ryuk ransomware is associated with cybercrimes because the large disruption can be used to force large organizations to pay ransoms to continue working.

This cyber-attack impacted daily operations and daily patient care. The hospitals under attack were forced to manually process patients. This new procedure included handwriting records, paper-based tracking, and face-to-face communication. During this time, the hospital experiences significant delays. The hospital would have to transfer patients and reschedule procedures. The doctors, not having access to official health records, compromised the care and the speed at which it was given throughout the hospital. Attacks like this can result in errors such as misdiagnoses and delays in care.

UnitedHealth Services reported that it did not pay the ransom requested by the attackers, and UnitedHealth Services was able to use backup systems to restore its systems. It took the UNS 3 weeks to fully recover their IT systems, costing them 67 million due to disruptions and recovery efforts, also damaging the organization's reputation.

How post-quantum cryptography can protect digital healthcare

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Lattice-based post cryptography is a cryptography method using algorithms based on mathematical structures called lattices. A lattice is points on a grid in a multi-dimensional space. These algorithms

are highly secure because of the difficulty of solving these algorithms. Traditional methods of encryption have been compromised by new techniques and quantum computers. Lattice-based encryption can overcome quantum attacks and is currently being standardized by the U.S. National Institute of Standards.

The United Health Services never confirmed exactly how the attackers gained access, implemented. Implementing lattice-based cryptography could have limited the access that the attackers had and reduced the chance of them stealing data or impersonating users. Lattice-based cryptography secures data even after it has been intercepted. Attackers can breach a network and download files and health records with traditional encryption, but with cryptography, the data would remain secure and unreadable even with a quantum computer if intercepted. This ability would be crucial for health facilities because health records must be protected for the duration of someone's life and possibly longer.

The most common way ransomware attacks occur is through phishing or using stolen credentials. This attack method allows the attacker to gain access to deploy malware in a system. Lattice-based cryptography digital signature schemes can protect systems from unauthorized access. These signature schemes verify that updates and authorization are legitimate sources. This means that even if an attacker gains access, they would be so limited in the network that they would be unable to deploy malware or impersonate users without being detected.

The 2021 incident at Universal Health Services demonstrated how cyberattacks can disrupt crucial infrastructure and endanger lives. Cyber-attacks will only become more sophisticated, and with the growing threat of quantum computers, organizations should implement updated security technologies such as lattice-based cryptography. Lattice-based cartography

offers a solution prepared for the future for protecting data and securing communication. When organizations like United Health Services adopt lattice-based cryptography, they will be able to defend against attacks that are currently being used and the attacks that will be used in the future, ensuring the safety of their operation and the people that trust them with their records.

VI. Protocol Analysis

A. Protocol 1: RSA

RSA is a foundational algorithm for modern cryptography security. RSA operates with public key encryption. RSA uses two pairs of keys: a public key for encryption and a private key for decryption. RSA can be used across most digital systems. This has helped RSA become a widely adopted method and is used for web browsers, email security, and many other applications. RSA is simple to manage and implement. Users can secure communication without having to authorize in advance if they have the key.

RSA remains secure against traditional computer attacks if the keys are configured correctly. The rise of quantum computing has begun to threaten the security of RSA. Shor's algorithms, which are used on quantum computers, could factor large numbers and break the core security of RSA. Another weakness of RSA is that it is not used for large dataset encryption. RSA key generation is relatively resource-intensive and can be slow. As computing power increases, RSA must use larger keys, which can place extra strain on resources and slow down the system.

B. Protocol 2: TLS/SSL

Transport Layer Security (TLS) is a cryptographic protocol. This protocol is designed to secure communication in a network. TLS protects data in transmission and ensures the data remains in its original state and has not been intercepted. The

major strength of TLS is its ability to encrypt sensitive information. TLS can encrypt passwords, credit card information, and messages. This protocol has been implemented in major web browsers and is protecting people all over the world right now.

TLS is a hybrid protocol that uses public key encryption and symmetric encryption for data transmission. The combination of both these methods has led to TLS being secure and practical for everyday use. Like many other key encryption methods, quantum computing can decrypt these keys, and data can be stolen because key encryptions will not remain protected like data protected by post-quantum cryptography.

TLS has document implementation problems. Misconfigurations and outdated versions have all been attacked in major security breaches. These events demonstrated how strong protocols can be weak if not maintained or configured properly, and researchers have begun the search for new hybrid methods, including quantum-resistant methods.

C. Protocol 3: Lattice-based encryption

Lattice-based encryption is the future of cryptography. This method of cryptography is designed to be effective against attacks from quantum computers. Lattices are hard mathematical problems that even quantum computers can break. The security of Lattice does not rely on factoring large numbers; this allows it to be immune to quantum computers and is the reason why the National Institute of Standards and Technology selected Lattice-based encryption for standardization.

Lattice-based encryption does come with its concerns. One of the concerns would be the size of the keys and ciphertext that are a challenge for low-power environments. Kyber requires large amounts of bandwidth and storage. Also, this is a new design and is still being researched. Lattice-based

encryption has not been working for multiple years, like other algorithms. Some researchers believe that there could be some unexpected vulnerabilities over time. Nevertheless, lattice-based encryption is expected to be the future of cryptography and protecting data from upcoming threats.

V. Case Study 3: The University of California System

In December 2020, the University of California (UC) system, a renowned network of higher education institutions, was targeted by a sophisticated cyberattack that exploited and compromised personal data within its supply chain due to a vulnerability in a third-party provider, Accellion which at the time suffered a nationwide attack that impacted over 100 companies (Twingate, 2024). As This attack resulted in a massive leak of confidential information from student academic records, including sensitive data such as dates of birth, education levels, genders, job titles, names, physical addresses, and social security numbers, which exposed them to malicious actors, affecting approximately half a million individuals. The attack employed advanced methods, including zero-day exploits and credential stuffing, which enabled unauthorized access to internal systems.

Zero-day exploits take advantage of undisclosed software vulnerabilities, which can often remain undetected, even to developers, at the time of the attack. Credential stuffing employs a brute-force method, where previously compromised credentials are used to hack into systems where users have reused their passwords. These techniques, combined, proved that traditional defenses had become inadequate, even with the UC system having implemented multi-factor authentication, Virtual Private Networks (VPNs), and Security Information and Event Management tools. (Trend Micro, 2021; Verizon DBIR, 2023).

While a large portion of the compromised information was encrypted, the threat to its long-term security became jeopardized due to the increasing threats posed by quantum computing. With the development of quantum computers on the rise, previously secure data could potentially be decrypted using one of quantum's more rare algorithms, such as Shor's algorithm. If run on a large enough quantum computer, this algorithm could break RSA by effectively finding its private and public keys. This would now compromise historical data, especially in Educational institutions where academic and research records need to be preserved for decades.(Shor, 1994; NIST, 2022).

Post-Quantum Cryptography (PQC) provides a system centered on building protection and safeguarding these assets by enabling these institutions to secure cloud backups and academic records against future decryption attempts. By integrating lattice-based algorithms and other quantum-resistant methods, PQC algorithms are designed to be resistant against both quantum and classical computers. In this sense PQC is designed not only to serve as a preventive measure but also as a long-lasting investment in data integrity.(Chen et al., 2016; NIST, 2022).

VII. PQC as a Double-Edged Sword

Although PQC offers a robust solution to the quantum threat, it introduces new complications. Early deployments of PQC are still under development and do not hold the same degree of maturity as classical encryption methods. If these underdeveloped systems are merged too soon or without proper validation, their data could be exploited.(Alkim et al., 2017; Bindel et al., 2019).

The transition to PQC typically includes hybrid models that combine both

quantum-resistant and classical algorithms. These transitional phases may unintentionally introduce security gaps, which become increasingly concerning for educational institutions that often face funding and resource constraints, potentially delaying rapid technological transformation.

Ethically and legally, the race to develop and implement PQC poses significant challenges. As nations and organizations race toward quantum readiness, discrepancies in capabilities could lead to unequal protection across sectors. Furthermore, the potential for offensive use of quantum tools raises questions about international norms and legal accountability in the quantum realm.

VIII. Recommendations

To prepare the educational sector for the quantum era, three key recommendations are proposed:

1) Sector-Specific Action Plans: Acts as a roadmap for incorporating national plans across various sectors like healthcare, transportation, and communication. Educational Institutions can benefit from developing clear PQC adoption frameworks created for their infrastructure. These plans are crucial for implementing a larger national strategy in an effective manner.(NIST NCCoE, 2022).

2) Hybrid Cryptographic Migration Strategies: A phased migration approach should be adopted using hybrid cryptographic systems. This ensures backward compatibility while gradually strengthening encryption standards.

3) Readiness Assessments and Training: Institutions must conduct periodic evaluations to assess their current cryptographic posture and identify potential gaps in their security. Staff and IT personnel should also receive training on PQC principles and quantum-related threats to

ensure knowledgeable adoption and management.

IX. Conclusion

The UC system breach serves as a critical reminder of the vulnerabilities facing the education sector. As traditional encryption becomes increasingly vulnerable in the face of quantum advancements, Post-Quantum Cryptography emerges as a vital line of defense. However, implementing PQC is not without risk. The technology must be integrated with caution, supported by sector-specific planning, hybrid strategies, and organizational awareness. With proactive preparation, the education sector can protect the integrity of academic records and research data, not just today, but for decades to come. (Mosca, 2018; NIST, 2022).

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